

Unusual Materials for Hydrogen Storage: From Natural Diamond to Inorganic Nanotubes and Inorganic Fullerene-Like Nanoparticles

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Low, Medium, and Large Scale Requirements
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Part 1: Study of Hydrogen in Diamond: Collaborators

Alon Hoffman (Technion, Israel)

Gérald Dujardin, Genevieve Comtet, (LPPM, Université Paris Sud, Orsay, France) – **Synchrotron Radiation Measurements**

Roger Azria, (LCAM, Université Paris Sud, Orsay, France) - **Electron Stimulated Desorption Measurements**

Presentation Outline

1. Exposure of differently oriented undoped natural diamonds (**type IIa-no impurities**) to deuterium activated by microwave (MW) plasma - SIMS analysis;
2. Interaction of activated by Microwave (MW) plasma and in UHV, by Hot Filament atomized deuterium with CVD polycrystalline diamond films – synchrotron radiation photon stimulated ion desorption (PSID) and electron stimulated desorption (ESD) measurements.

Aim of the Research

1. To elucidate, whether hydrogen/deuterium can diffuse into the inner lying atomic layers of natural undoped diamond following exposure to MW plasma.
2. To compare the effects of MW and *in situ* (in UHV) deuteration (using hot tungsten filament) on the structure and electronic properties of diamond.

Interaction of MW Plasma Activated Deuterium with Natural Diamond

Sample Preparation

Samples: undoped, type IIa, natural (100)-, (110)-, and (111)-oriented diamonds

Treatment: exposure to MW Deuterium (99.999 %) plasma:

Gas pressure: 60 Torr;

Sample Temperature: ~ 700 °C

Plasma Power: 900 W

Duration of the treatment: 30-45 min

Analysis

Static SIMS: surface concentration of deuterium

Dynamic SIMS: sputter depth profile

Verification of the Crystalline Perfection

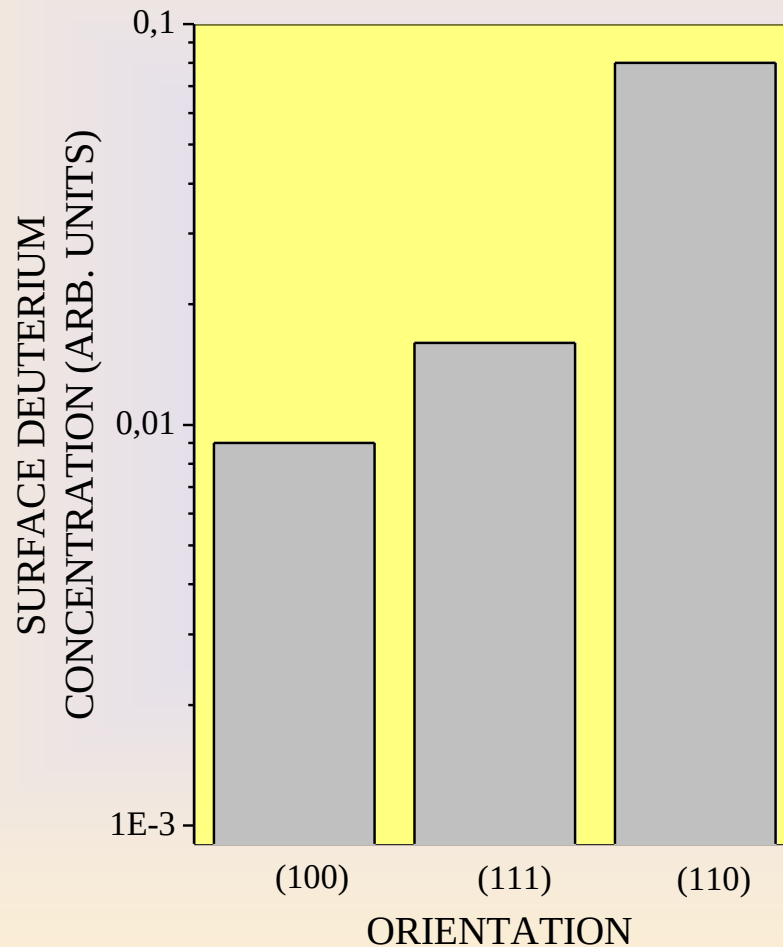
(before and after MW plasma treatment)

LEED pattern: well-defined planes: (111)-1×1:D, (100)-2×1:D, and (111)-facetted structure for the (110)-oriented diamond

EELS: the characteristic diamond bulk and surface plasmon peaks at 33 and 23 eV, respectively, were observed with no traces of spectral features at 6 eV commonly related to amorphous carbon and graphite

Static SIMS

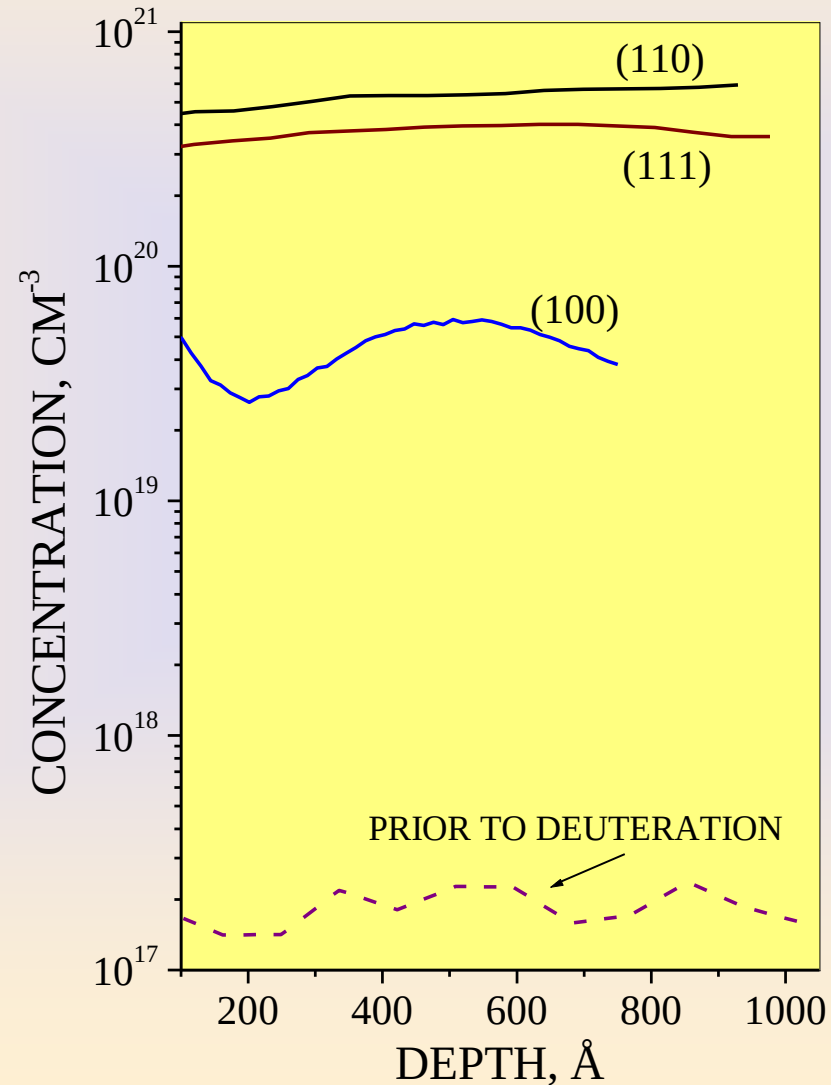
Experimental: 10 keV Cs⁺ beam
Ion current: 1×10^{-12} A
Maximal ion dose: 4×10^{12} cm⁻²



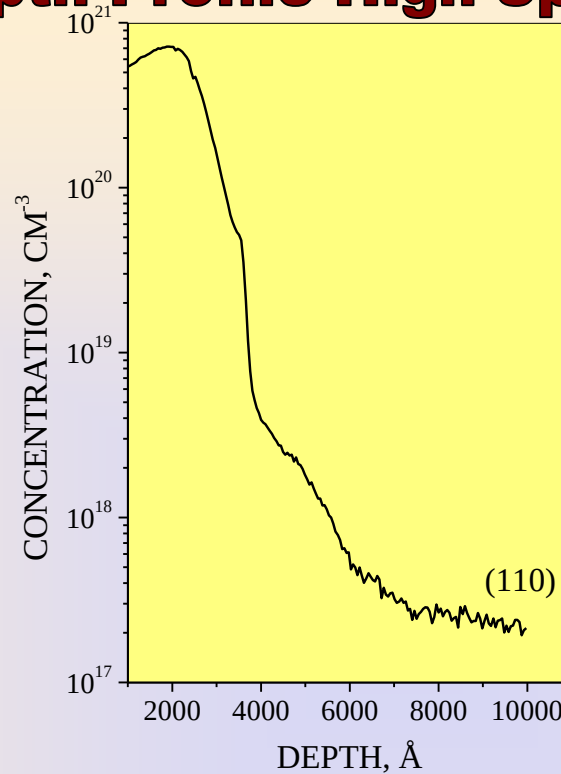
The highest concentration of deuterium is on the **(110)-oriented** surface.

Depth Profile SIMS Analysis

Experimental: 10 keV Cs⁺ beam;
Ion current: 1×10^{-8} A



Depth Profile-High Sputter Rate



- Deuterium atoms activated by MW plasma **diffuse** into undoped type IIa diamond crystal to doses of 10¹⁹-10²¹ cm⁻³
- **No change** in hydrogen concentration **when hydrogenated samples have been stored for days or weeks**
- Relative amounts of deuterium in the **bulk**, as well as in the **near-surface** region of differently oriented diamonds **retain these on the very surface**:
 - 10²⁰-10²¹ cm⁻³ for the (110)-oriented surface
 - ~ 10²⁰ cm⁻³ for the (111)-oriented surface
 - ~ 10¹⁹ cm⁻³ for the (100)-oriented diamond
- The diffusion depth of deuterium is ~ 8000 Å (0.8 μm)

Interaction of Activated Deuterium with Polycrystalline Diamond Films: MW Plasma vs. Hot Filament Activation

Sample Preparation

Substrates : *p*-doped Si wafers, treated with diamond/ethanol slurry

Deposition: using MW CVD

CH₄:H₂ (1.5:98.5) gas mixture

Gas pressure: 70 Torr (400 sccm gas flow rate)

Temperature: ~ 850 °C (MW power: 1200 W)

Deposition time: 4.5 hours

Film Thickness: ~ 10 μm

Crystallite Size: 2-5 μm

Post-Deposition Deuteration (*Ex situ*):

exposure to MW Deuterium (99.999 %) plasma:

- ✧ Gas pressure: 60 Torr
- ✧ Sample Temperature: ~ 800 °C
- ✧ Plasma Power: 900 W
- ✧ Duration of the treatment: 30-45 min

Deuteration (*In situ*):

- ✧ Annealing of the film to 1000 °C in UHV
- ✧ Verification of the complete hydrogen desorption (by PSID)
- ✧ Exposure to deuterium gas flowing through a hot (1800 °C) tungsten filament, while keeping the sample at the room temperature:
 - ☞ Gas Pressure: 1.5×10^{-5} Torr (while pumping)
 - ☞ Duration: 60 min

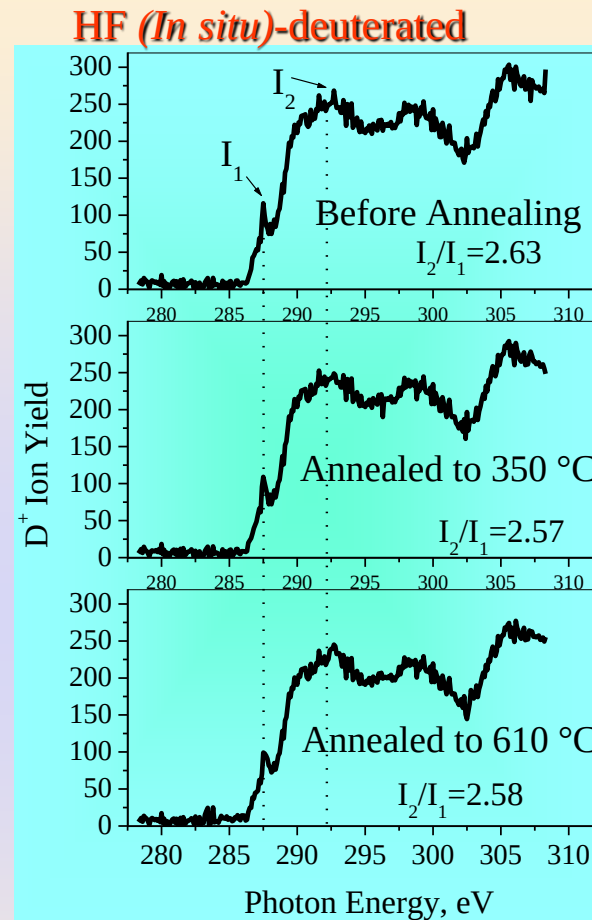
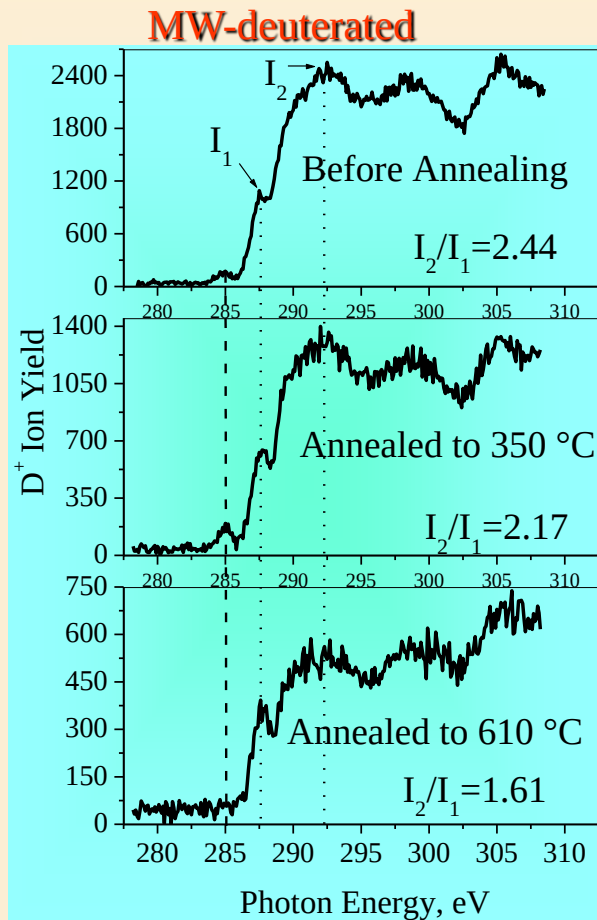
Analysis

SIMS measurements - *difficult to perform* due to high surface roughness

We applied the following techniques:

1. **Photon stimulated ion desorption (PSID) of D^+ :**
using synchrotron radiation: 280-310 eV
measuring the total yield of escaping D^+ ions after *in situ* annealing to 25–1000 °C
2. **Electron stimulated desorption (ESD) of D^- (H^-):**
using low energy electrons (5–35 eV)
measuring:
 - ↳ ion yield
 - ↳ ion kinetic energy distribution (KED)

PSID measurements



Assignment of the spectral features: [Reference: *Hoffman et al.*, Phys. Rev. B **62**, 8446 (2000)].

I_1 (287.5eV)-excitation energy of C-H (ads) bonds: *direct* (via core-hole relaxation) and *indirect* (via secondary electron excitations) processes;

I_2 (~ 292 eV) - due to excitation of C-C bonds (mainly *indirect* processes);

285 eV – valence band excitations due to *sec. electrons* in sp^2 -bonded carbon;

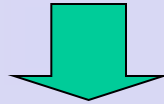
302.4 eV (*dip*) – second absolute band gap of diamond (*indirect* processes).

MW-deuterated films: effect of annealing:

- ☞ integral desorption yield *decreases*
- ☞ I_2/I_1 peaks ratio *decreases*

HF (*in situ*)-deuterated films: effect of annealing:

- ☞ integral desorption yield *is lower* than for a MW-deuterated film
- ☞ integral desorption yield *remains nearly constant*
- ☞ I_2/I_1 peaks ratio *remains nearly constant*



MW-deuterated films:

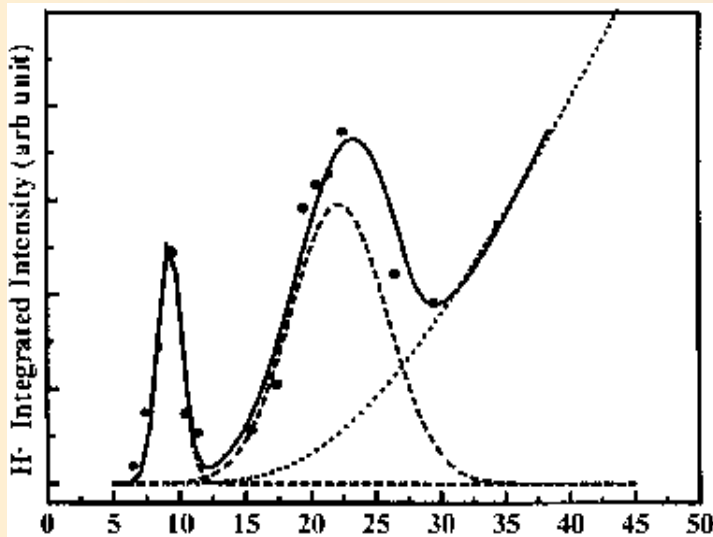
- ☞ deuterium concentration decreases upon moderate heating
- ☞ dynamics of the desorption changes in favor of direct processes (*not driven by SEE*)

HF (*in situ*)-deuterated films:

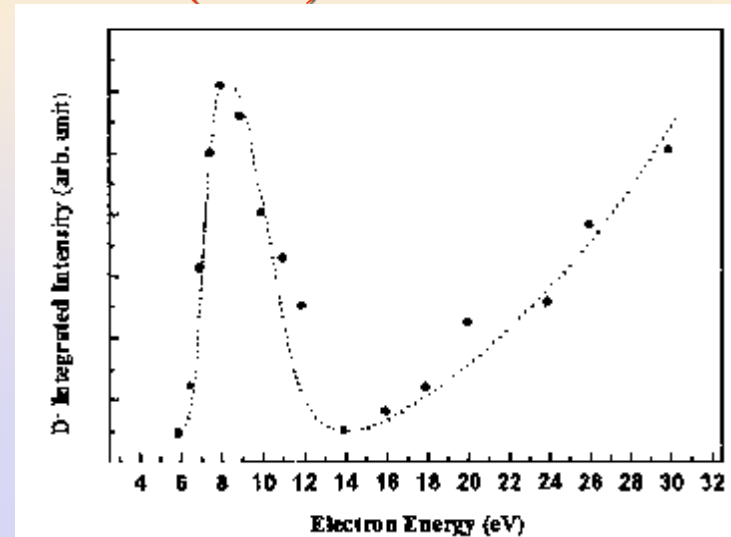
- ☞ deuterium is stable upon annealing
- ☞ dynamics of the desorption processes remains unchanged

ESD measurements

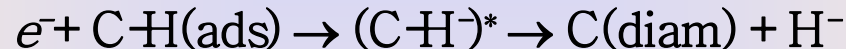
MW-deuterated



HF (*in situ*)-deuterated



✧ The peak around 9 eV is attributed to a dissociative electron attachment (DEA) process through a Feshbach resonance:



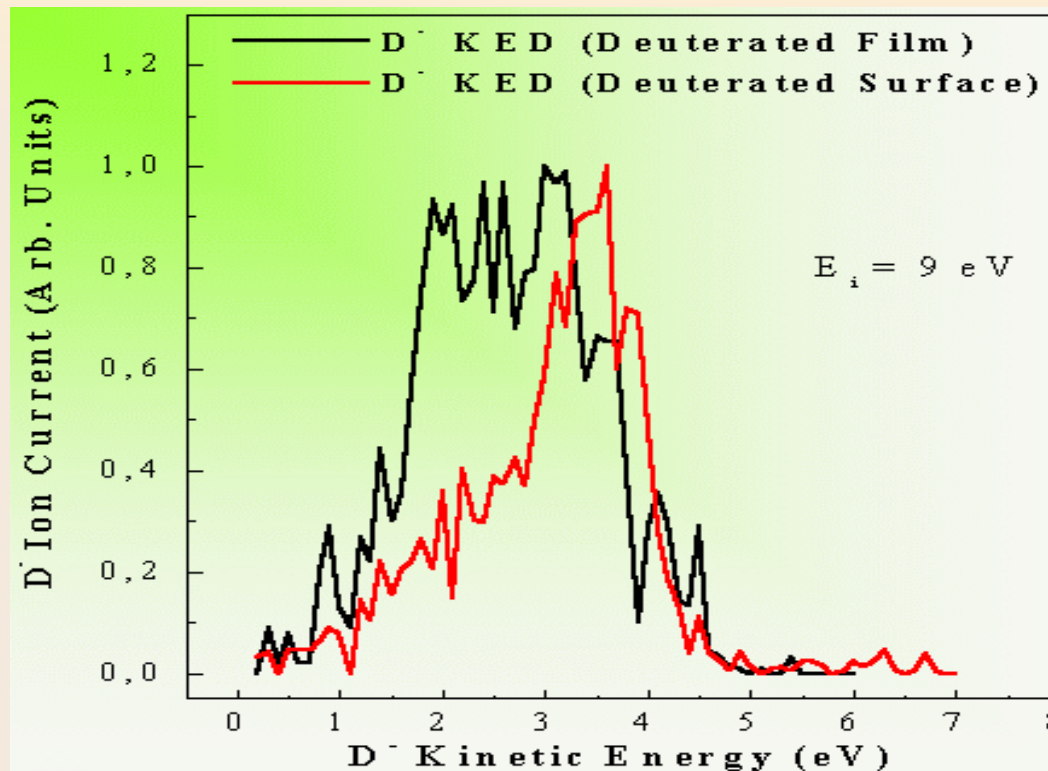
✧ The monotonic increase is associated with a dipolar dissociation (DD) process:



✧ The structure around 22 eV is due to sub-surface deuterium giving DEA process after the incident electrons have lost energy (13 eV) in electronic inter-band excitation of the diamond

Reference: Hoffman *et al.*, Phys. Rev. B **63**, 045401 (2001).

Kinetic Energy Distribution of Deuterium Anions



- ✧ In the «surface» case (HF-deuterated), the D^- KED displays a peak at a high kinetic energy and a low-intensity tail
- ✧ In the «film» case (MW-deuterated), deuterium is located in the sub-surface, as well as on the surface itself. This results in energy losses due to *collisions and phonon coupling before D^- ions escape from the film*

Hydrogen in Diamond: Conclusions

Natural Diamond:

- ↻ Deuterium activated by MW plasma diffuses into undoped type IIa natural diamond to the concentrations of 10^{19} – 10^{21} cm⁻³
- ↻ Both bulk and surface concentrations of deuterium are the highest for the (110)-oriented surface, and the lowest for the (100)-oriented one. D
- ↻ Diffusion depth of deuterium is in the order of 0.8 μm
- ↻ Hydrogen can be stored for long periods (days, weeks, may be moore)

Polycrystalline Diamond Films:

- ↻ Deuteration performed *in situ* (by Hot Filament) does not result in bulk diffusion of deuterium atoms, in contrast to the MW plasma treatment
- ↻ Bulk deuterium (hydrogen) is not thermally stable, and desorbs upon moderate heating to 350 °C
- ↻ Chemisorbed deuterium (hydrogen) atoms are stable up to high temperatures.

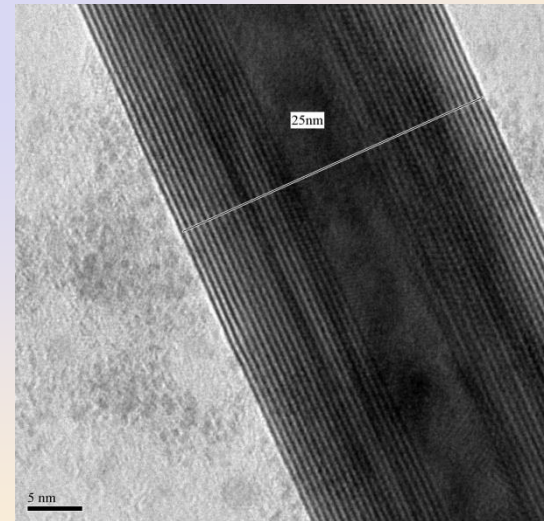
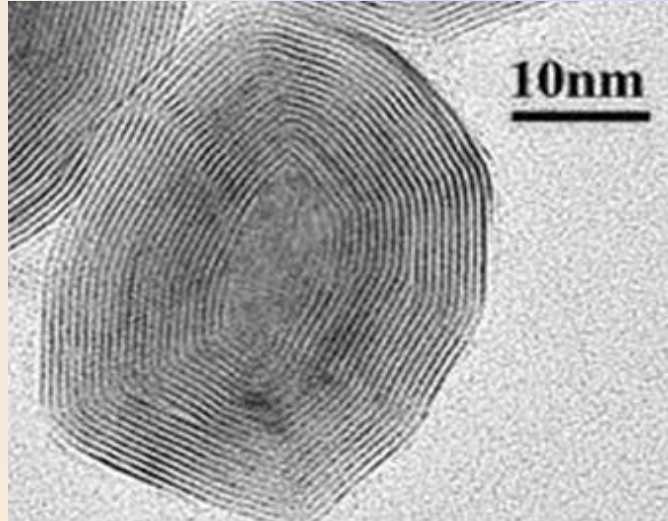
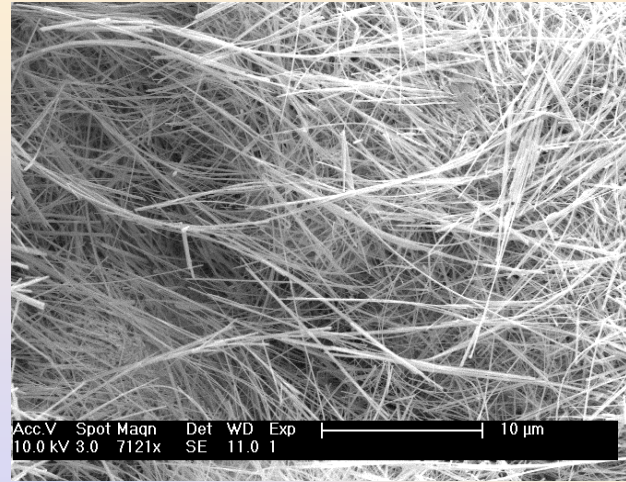
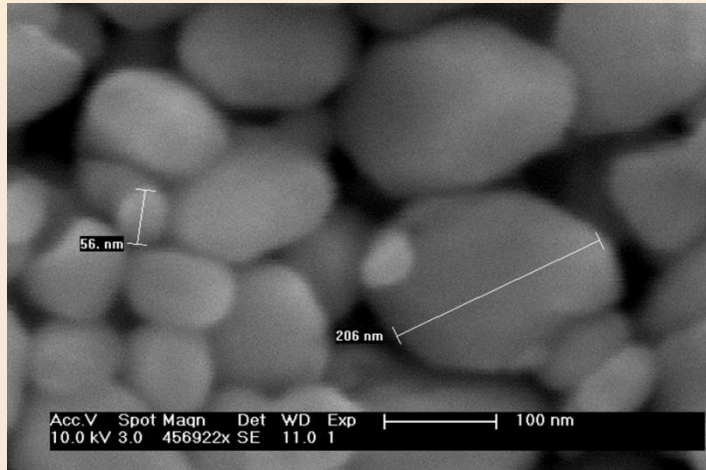
Study of Hydrogen in Tungsten Sulfide Nanoparticles: Collaborators

Alla Zak – HIT and Weizmann Institute, Israel – synthesis

Shaul Michaelson, Alon Hoffman - Technion, Israel - hydrogenations

Hoi Ri Moon – UNIST, Korea – volumetric measurements

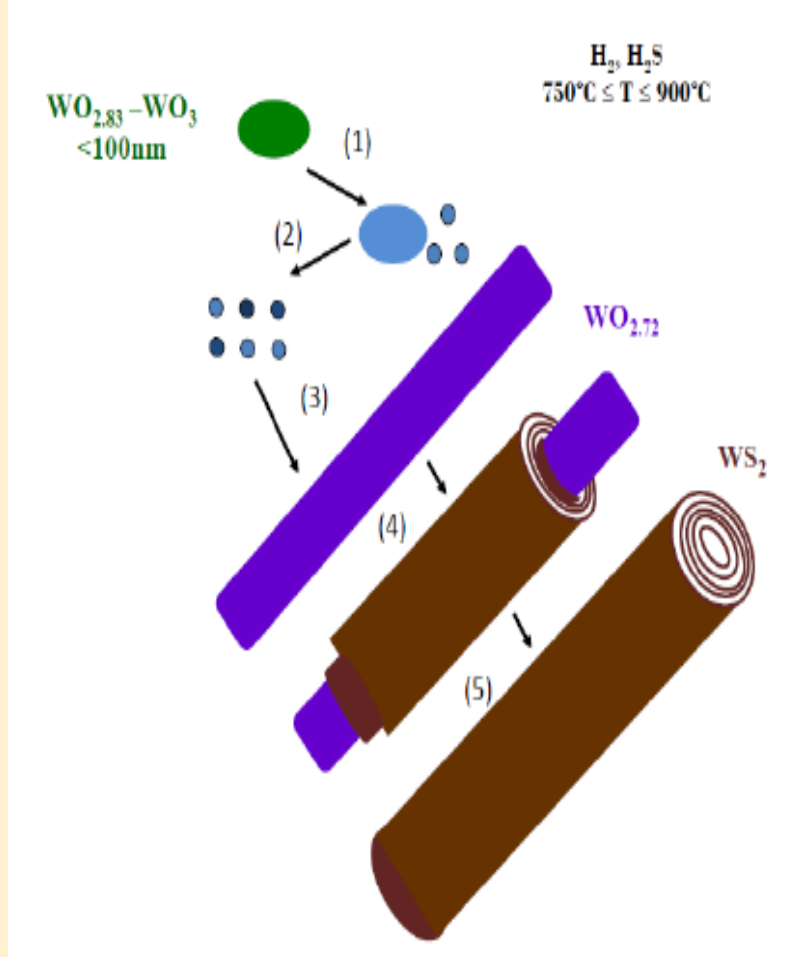
The Substrate Materials



WS₂-Inorganic Fullerenes
(IF)

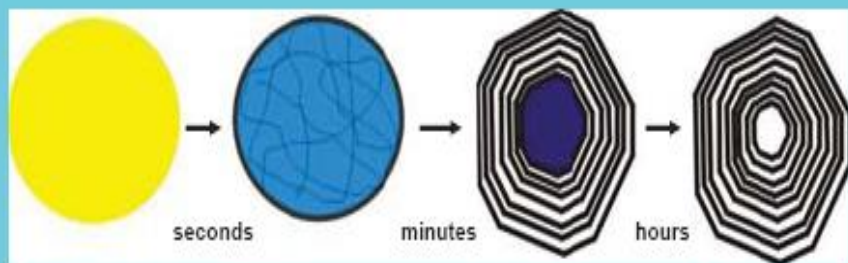
WS₂-Inorganic Nanotubes
(INT)

How to Prepare?



1. The nanosize precursor mixture of non-volatile oxide $\text{WO}_{2.83}\text{--WO}_3$ phases undergoes mild reduction by hydrogen gas producing a volatile oxide phase (with $2.72 < \text{O/W} < 2.83$).
2. The oxide vapor undergoes additional partial reduction to the non-volatile phase, which serves as a nucleation center for the nanowhisker growth into stable suboxide phase of $\text{WO}_{2.72}$ ($\text{W}_{18}\text{O}_{49}$).
3. Sulfidization starts from the surface and middle of the oxide nanowhiskers, and oxygen to sulfur exchange progresses towards the end of the nanowhiskers (fast step) and inwards (slow-diffusion controlled reaction). This process leads to a gradual consumption of the oxide core creating a hollow multiwall WS_2 NT.

Conversion of the WO_3 oxide nanoparticles into inorganic fullerene like nanoparticles of WS_2



Precursor: WO_3 nanoparticle

the WO_3 suboxide nanoparticle, covered with few layers of disulfide

WS_2 closed-cage structure with a suboxide core encapsulated

Fully converted WS_2 inorganic fullerene-like nanoparticle

➤ Fast reaction of the H_2S and H_2 with the surface of semi-spherical oxide NP (the precursor) leads to the quick formation of a few closed WS_2 layers encapsulating the oxide core.

➤ The surface-passivation of the oxide NP by the inert layers of WS_2 prevents coarsening of the NP, making each one of them a nanoreactor of its own. Thus, the oxide core becomes partially reduced to WO_{3-x} and, subsequently, a slow diffusion-controlled reaction leads to fully replacement of the oxygen with sulfur atoms.

➤ This outwards-inwards-reaction proceeds in epitaxial-like fashion (layer by layer, using the external sulfide layers as a template) until the synthesis of the hollow IF- WS_2 NP is completed.

➤ The size and shape of the fullerene-like NP (IF- WS_2) is determined by the size and shape of the pre-prepared WO_3 NP. The majority of oxide/IF NP range from **150 to 350 nm** in diameter being of oval shape.

➤ The presence of a hollow core in the IF NP is attributed to the differences in specific gravity of WS_2 and WO_3 (7.5g/cm^3 and 7.15g/cm^3 , correspondingly).

Why to use them for hydrogen storage?

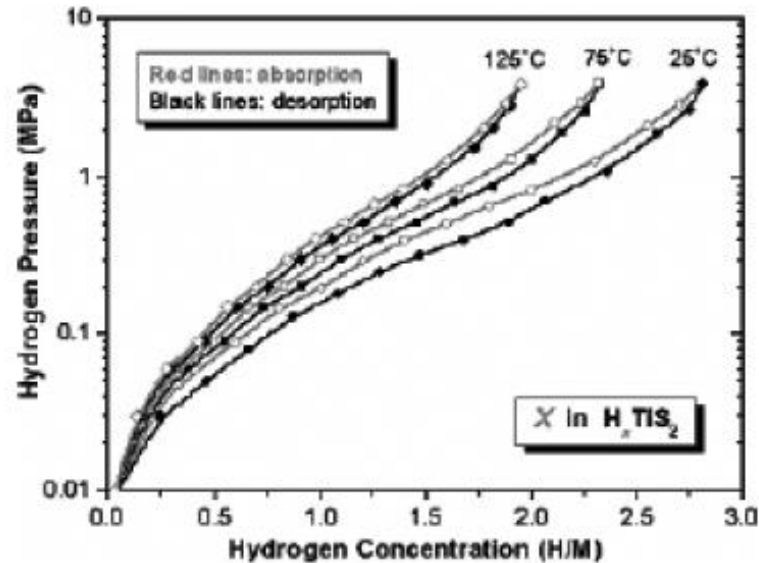
- High thermal and mechanical stability
- Layered and packed structure
- H₂ could be intercalated in between the layers
- H₂ can diffuse into hollow core or in between the layers through surface defects of IF spherical particles, and in the case of INT – through the open end of the tubes

How to treat them to store hydrogen?

- To expose to molecular hydrogen at high pressure, high/low temperature
- To expose to plasma-activated (MW, RF, DC) hydrogen

Early Works on Hydrogen Storage in XS_2 Nanotubes

TiS_2

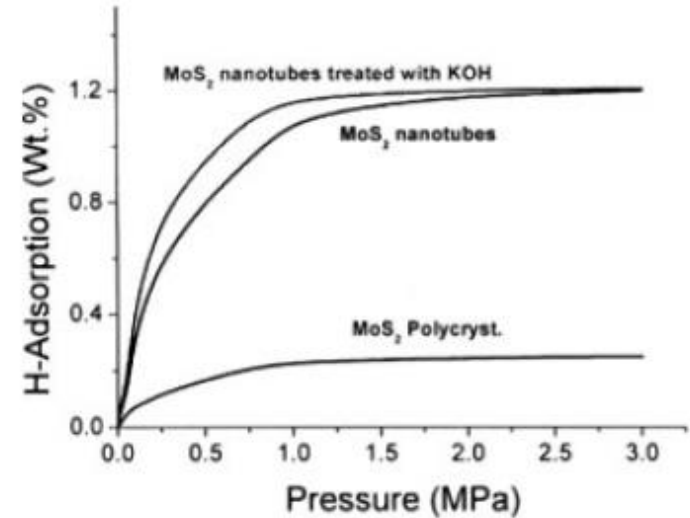


J. Chen, S.-L. Li, Z.-L. Tao, Y.-T. Shen, C.-X. Cui, *J. Am. Chem. Soc.* **2003**, 125, 5284.

Max. Wt.%: 2.8%

Molecular weight, TiS_2 : 112 g/mol

MoS_2



J. Chen, S. L. Li, Z. L. Tao, *J. Alloys Compd.* **2003**, 356–357, 413.

Max. Wt.%: 1.2%

Molecular weight, MoS_2 : 160 g/mol

Recent Results:

Exposure of WS₂-INT to High Pressure Molecular Hydrogen

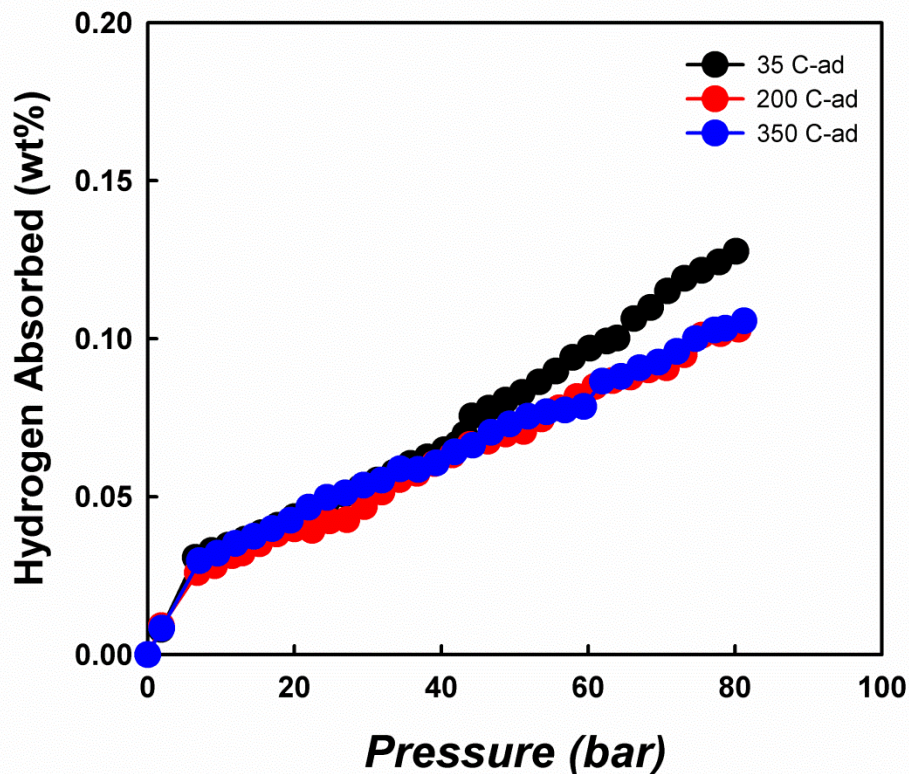
Outgassing conditions: 120 °C annealing under vacuum for 8 h

After each isotherm: 350 °C annealing under vacuum for 8 h

Adsorption time: 10 min

20 sec (time interval between data collection)

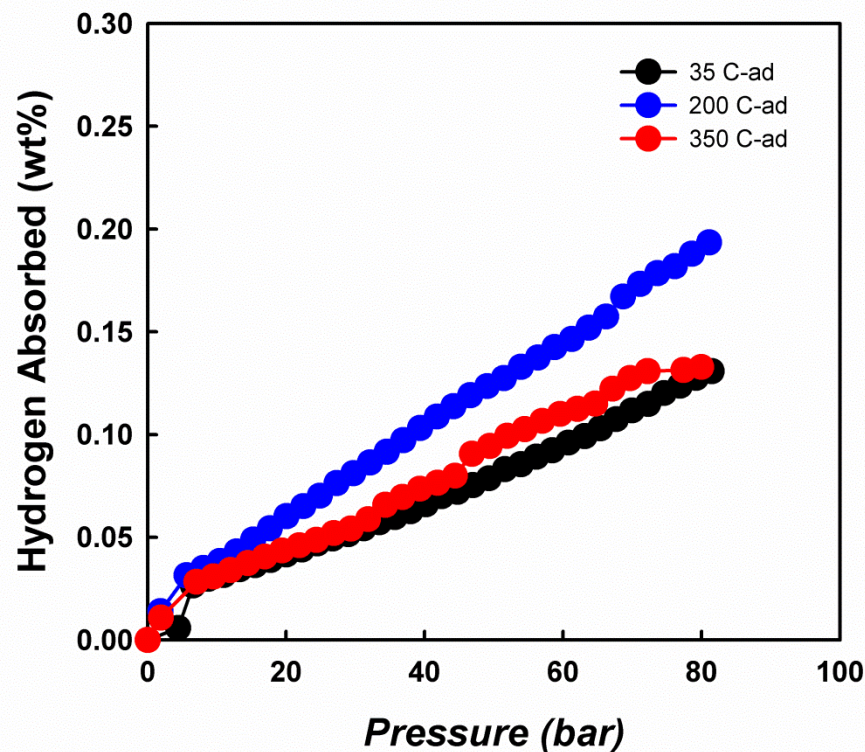
Verification of the substrate material: XRD and SEM



adsorption temperatur e	wt%
35 °C	0.13
200 °C	0.10
350 °C	0.11

Recent Results:

Exposure of WS₂-IF to High Pressure Molecular Hydrogen



adsorption temperatur e	wt%
35 °C	0.13
200 °C	0.19
350 °C	0.13

M.W. (WS₂) = 248 g/mol

Recent results: Exposure of WS₂-INT to Low Pressure MW-Plasma Activated Hydrogen

Goal: to get a sufficient energy (per an activated gas molecule)
and large atomic flux to the substrate!

Conditions:

400 W, 60 Torr, 200 sccm, 5 min

Sample: tablet of 5 mm in diameter

$$Flux = \frac{6 \times 10^{23} \text{ mol}^{-1}}{22.4 \times 10^{-3} \text{ m}^3 / \text{mol}} \cdot v_T \cdot A \cdot t = 2.5 \times 10^{25} \times 1500 \cdot 2 \times 10^{-5} \cdot 300 \cdot \frac{60 \text{ torr}}{760 \text{ torr}} = 1.5 \times 10^{25} (\text{atoms}) (\text{eq. 1})$$

And the energy per an atom is (Eq. 2):

$$Power = \frac{400}{1.67 \times 10^{-19}} = 2.5 \times 10^{21} \left(\frac{\text{eV}}{\text{s}} \right); N/t = \frac{200 \text{ sccm}}{22400 \text{ cc/mol}} \cdot \frac{6 \times 10^{23} \text{ mol}^{-1}}{60 \text{ s/min}} \approx 1 \times 10^{20} \frac{\text{atoms}}{\text{s}}$$

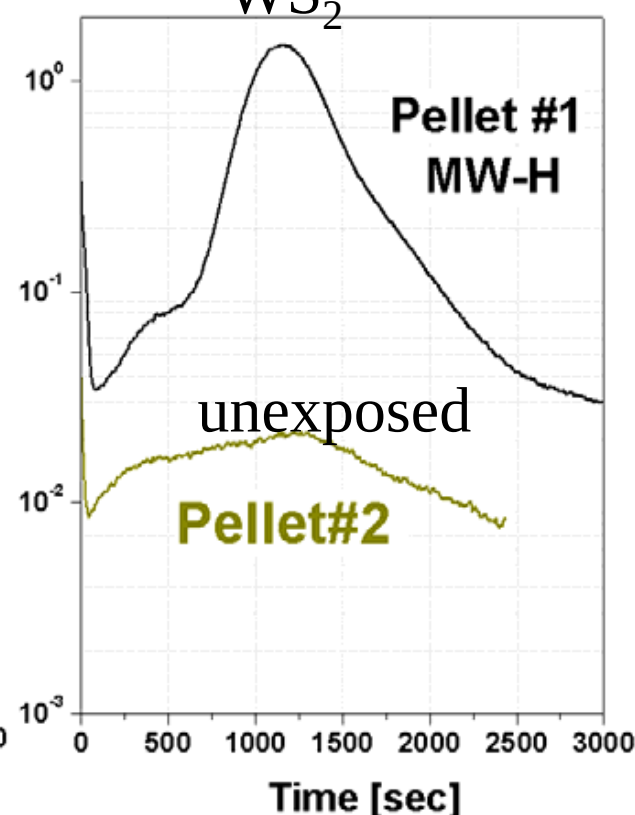
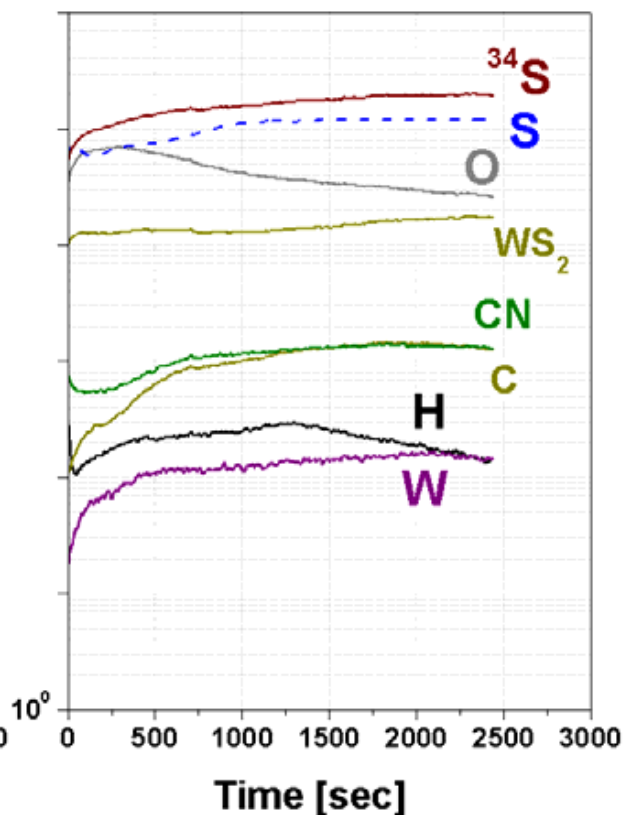
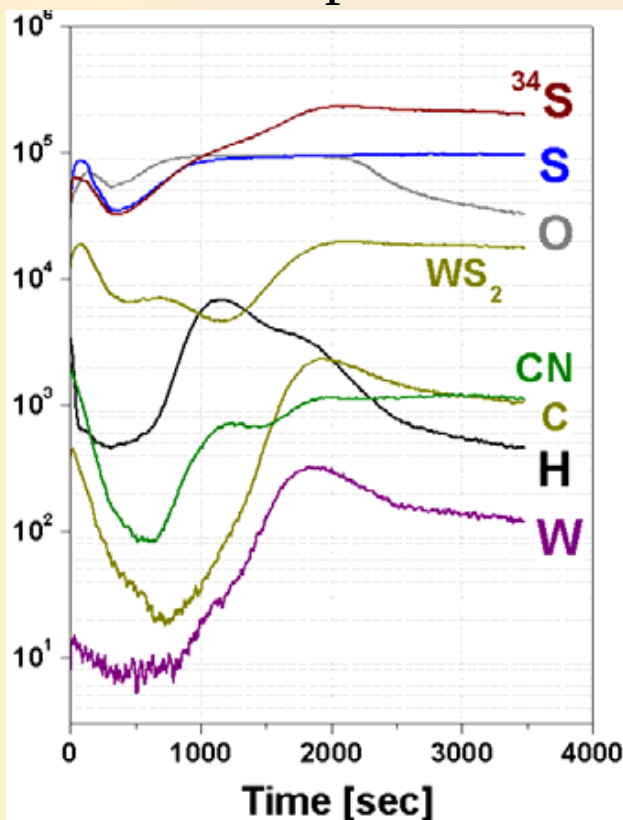
$$Energy = \frac{2.5 \times 10^{21}}{1 \times 10^{20}} = 25 \left(\frac{\text{eV}}{\text{atom}} \right) (\text{eq. 2})$$

TOF SIMS for hydrogen detection

MW-exposed

Unexposed

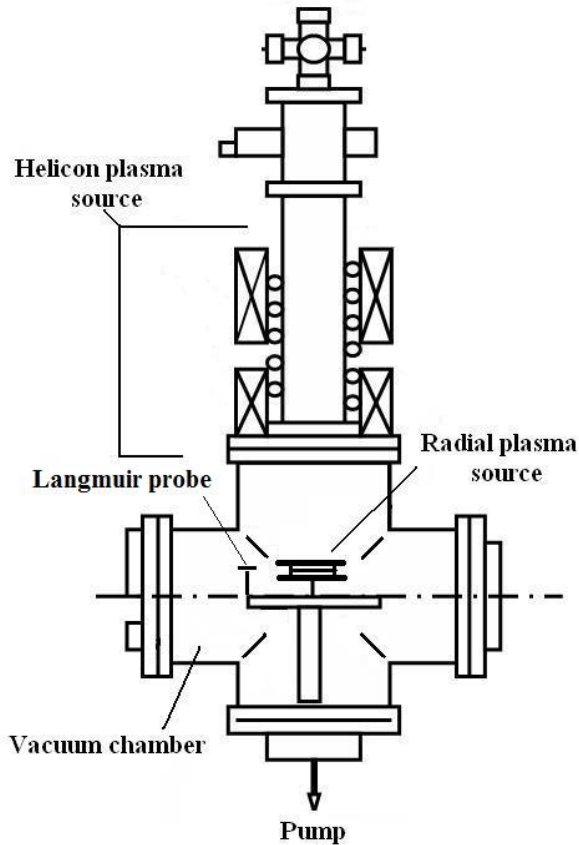
Both: normalized to WS_2



SEM verified: no visible damage to the structure

- Hydrogen signal increase: **x70** (as related to the unexposed sample)
- Hydrogen concentration in the unexposed sample: ~ 0.015 wt. %
- Estimated hydrogen concentration in the MW-exposed WS_2 -INT: **~ 1 wt. %:**
to be verified by XRD and PCT hydrogen desorption

The Ongoing Research: RF/DC Plasma: Testing the Influence of Particles Energy and Momentum



- Different gas flow rate and pressure
- RF plasma: better control on the energy of particles in plasma
- DC plasma: better control over the momentum of particles

In Situ Diagnostics of Hydrogen Adsorption/Desorption in the Plasma System

- After hydrogenation the vacuum chamber will be pumped down to $\sim 10^{-5}$ Torr.
- The sample will be heated to ~ 500 °C.
- The hydrogen desorption will be monitored by measuring the gas pressure versus the time, $P(t)$.
- Using data of $P(t)$ and the pumping speed data $S(P)$, we will calculate the mass of the gas absorbed by the sample as $m = M \int_0^t S \frac{P}{kT} dt - m_0$, where m_0 is the mass of the gas absorbed by experimental setup and M is the molecular mass of hydrogen.

Hydrogen in WS₂-INT and WS₂-IF: Conclusions

- Molecular high pressure-room/high temperature hydrogen absorbs into WS₂-INT and WS₂-IF to modest 0.13-0.19 wt.%.
- Increasing temperature does not really favor hydrogen absorption.
- Very short (5 min) MW-plasma activated hydrogen exposure results in up to ~1 wt.% absorption of hydrogen in WS₂-INT (rough estimation).
- Multiple cycles of exposure-release can be performed and no deterioration of the substrate material has been detected.

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